

Question	Answer	Marks
5(a)	(loss in) kinetic energy of α -particle = $Qq / 4\pi\epsilon_0 r$ or $7.7 \times 10^{-13} = Qq / 4\pi\epsilon_0 r$	C1
	$7.7 \times 10^{-13} = 8.99 \times 10^9 \times 79 \times 2 \times (1.60 \times 10^{-19})^2 / r$	M1
	$r = 4.7 \times 10^{-14} \text{ m}$ r is closest distance of approach so radius less than this	A1
5(b)	force = $Qq / 4\pi\epsilon_0 r^2 = 4u \times a$	C1
	$8.99 \times 10^9 \times 79 \times 2 \times (1.60 \times 10^{-19})^2 / (4.7 \times 10^{-14})^2 = 4 \times 1.66 \times 10^{-27} \times a$	C1
	$a = 2.5 \times 10^{27} \text{ m s}^{-2}$	A1
5(c)	so that single interactions between nucleus and α -particle can be studied or so that multiple deflections with nucleus do not occur	B1

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C(c)(i)	lamp needs 'high' power/'large' current/'large' voltage	D1